

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – Data Interpretation

Course Code:	DSA03	Course Title:	Natural Language Processing
CO Assessed:	CO4 – Precision	Assessment:	Assessment Tool 1 – Data Interpretation
Weightage:	35% of CIA		
CO5 Statement:	Formulate context-free grammars, feature structures, probabilistic parsing techniques and to construct parsing frameworks. (BL-4)		
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Section / Batch:	BTECH AI & DS	Date:	12-08-2026

Code of Conduct

/ Renee Vincent ASR-192424161 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Renee Vincent

Instructions

- Read each data set, table, semantic representation, or scenario carefully before answering.
- Analyze the given information and provide logical, evidence-based interpretations.
- Show all intermediate reasoning, semantic analysis steps, predicate representations, or calculations wherever applicable.
- Answers should be concise, structured, and technically accurate.
- Support your conclusions using the data provided in the question.
- Draw diagrams, semantic networks, parse trees, or logical representations wherever relevant.
- Avoid copying definitions alone; focus on analyzing the given scenario and interpreting the data.
- Duration: 30 minutes.

Section / Topic	Focus	Question No.
Representing Meaning & Meaning Structure of Language	Analyze semantic representations, identify action–entity relationships, detect interpretation errors, and evaluate meaning representation in chatbot systems.	Q1
First-Order Predicate Calculus & Representing Linguistically Relevant Concepts	Construct predicate representations, apply logical inference rules, derive conclusions, and evaluate reasoning in smart manufacturing environments.	Q2
Lexemes and Their Senses & Word Sense Disambiguation	Analyze contextual clues, determine correct word senses, evaluate ambiguity resolution, and improve semantic understanding in search systems.	Q3
Syntax-Driven Semantic Analysis	Analyze syntactic structures, assign semantic roles, evaluate parsing accuracy, and improve semantic interpretation in healthcare NLP applications.	Q4

Annexure A – Data Interpretation

1. Semantic Representation in Customer Support Chatbots.
A telecom company collected customer chatbot interactions. The semantic analyzer generated the following meaning representations.

Customer Query	Semantic Representation
"Activate international roaming for my number."	ACTIVATE(Roaming, Customer)
"Deactivate caller tune service."	DEACTIVATE(CallerTune, Customer)
"Check my data balance."	QUERY(DataBalance, Customer)
"Enable 5G service."	ACTIVATE(5GService, Customer)

Additional chatbot logs show:

Query ID	Actual User Intent	Predicted Intent
Q1	Activate Roaming	Activate Roaming
Q2	Deactivate Caller Tune	Activate Caller Tune
Q3	Check Data Balance	Query Data Balance
Q4	Enable 5G Service	Activate 5G Service

Tasks:

- Analyze the semantic representations and identify the action-object relationships.
 - Determine which query contains semantic interpretation errors.
 - Evaluate how meaning representation affects chatbot decision accuracy.
 - Recommend improvements to enhance semantic understanding in telecom customer support systems.
2. First-Order Predicate Calculus for Smart Manufacturing.
An Industry 4.0 manufacturing plant monitors machines using semantic rules.
Production Data:

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

Predicate Rules:

- $\text{Active}(x) \rightarrow \text{Producing}(x)$
- $\text{Produces}(x,y) \wedge \text{Active}(x) \rightarrow \text{Available}(y)$
- $\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)$

Tasks:

- Represent the production data using First-Order Predicate Calculus.
 - Analyze which products are currently available.
 - Infer whether Gear production is affected by maintenance activities.
 - Evaluate the effectiveness of predicate logic in industrial decision-support systems.
3. Word Sense Disambiguation in E-Commerce Search Engines.
An e-commerce company observed the following search logs.

User Query	Possible Sense 1	Possible Sense 2
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"Apple accessories"	Fruit	Technology Brand
"Mouse wireless"	Animal	Computer Device
"Java tutorial"	Island	Programming Language
"Python course"	Snake	Programming Language

Search Context Data:

Query	Clicked Result
Apple accessories	iPhone Charger
Mouse wireless	Bluetooth Mouse
Java tutorial	Coding Lessons
Python course	Software Development Training

Tasks:

1. Analyze the contextual information and determine the correct word sense for each query.
2. Identify the semantic cues used for disambiguation.
3. Evaluate the impact of incorrect sense selection on search engine performance.
4. Suggest an industrial-scale WSD strategy for improving recommendation accuracy.

4. Syntax-Driven Semantic Analysis in Healthcare Information Systems.

A healthcare NLP system processes medical reports.

Parsed Sentences:

Sentence	Parse Structure
Doctor prescribed medicine to patient.	Subject-Verb-Object
Patient reported severe headache.	Subject-Verb-Object
Nurse monitored patient continuously.	Subject-Verb-Object
Medicine reduced blood pressure.	Subject-Verb-Object

Semantic Roles Generated:

Entity	Role
Doctor	Agent
Medicine	Instrument
Patient	Recipient
Headache	Symptom

Tasks:

1. Analyze how syntactic structures contribute to semantic role assignment.
2. Determine whether the assigned semantic roles are appropriate.
3. Evaluate potential errors that could arise from incorrect parsing.
4. Recommend methods to improve syntax-driven semantic analysis in healthcare applications.

CASE STUDY 1: SEMANTIC REPRESENTATION IN CUSTOMER SUPPORT CHATBOTS

1. Action-Object Relationships

The semantic representations follow the structure:

ACTION(Object, Customer)

Customer Query	Action	Object	Meaning
Activate international roaming for my number.	ACTIVATE	Roaming	Customer wants to activate roaming
Deactivate caller tune service.	DEACTIVATE	CallerTune	Customer wants to deactivate caller tune
Check my data balance.	QUERY	DataBalance	Customer wants to check data balance
Enable 5G service.	ACTIVATE	5GService	Customer wants to activate 5G

Analysis

- **ACTIVATE(Roaming, Customer)** → Activate roaming service.
- **DEACTIVATE(CallerTune, Customer)** → Deactivate caller tune service.
- **QUERY(DataBalance, Customer)** → Retrieve the customer's data balance.
- **ACTIVATE(5GService, Customer)** → Enable 5G service.

The **action** identifies what the customer wants to do, while the **object** identifies the telecom service affected by that action.

2. Semantic Interpretation Errors

Given Results

Query ID	Actual Intent	Predicted Intent	Status
Q1	Activate Roaming	Activate Roaming	Correct
Q2	Deactivate Caller Tune	Activate Caller Tune	Incorrect
Q3	Check Data Balance	Query Data Balance	Correct
Q4	Enable 5G Service	Activate 5G Service	Correct

Analysis

The semantic interpretation error occurs in **Q2**.

Actual intent:

DEACTIVATE(CallerTune, Customer)

Predicted intent:

ACTIVATE(CallerTune, Customer)

The system correctly identifies the object **CallerTune**, but incorrectly identifies the action as **ACTIVATE** instead of **DEACTIVATE**.

Therefore, the error is specifically in **action identification**, not object identification.

Accuracy

Total queries = 4

Correct predictions = 3

Incorrect predictions = 1

Accuracy:

$\text{Accuracy} = (\text{Correct Predictions} / \text{Total Predictions}) \times 100$

$\text{Accuracy} = (3 / 4) \times 100$

Accuracy = **75%**

Therefore, the chatbot achieves **75% intent prediction accuracy** on the given examples.

3. Effect of Meaning Representation

Meaning representation converts natural-language queries into structured semantic forms such as:

ACTIVATE(Roaming, Customer)

This representation helps the chatbot separate the customer's **action** from the **service or object** involved.

For example:

"Activate international roaming for my number."

is represented as:

ACTIVATE(Roaming, Customer)

This allows the system to identify:

- Action = ACTIVATE
- Object = Roaming
- Entity = Customer

Similarly:

"Deactivate caller tune service."

becomes:

DEACTIVATE(CallerTune, Customer)

Impact on Decision Accuracy

Accurate semantic representation improves:

- Intent classification.
- Service identification.

- Action selection.
- Response generation.
- Backend service execution.
- Customer satisfaction.

However, an incorrect action representation can lead to an incorrect business operation. In Q2, interpreting **DEACTIVATE** as **ACTIVATE** could cause the chatbot to perform the opposite operation requested by the customer.

Therefore, semantic analysis is critical in telecom systems because incorrect interpretation can result in incorrect service activation or deactivation.

4. Recommended Improvements

1. Context-Aware Analysis

The system should consider the complete sentence rather than relying only on individual keywords.

For example:

"Deactivate caller tune service."

must be interpreted as a deactivation request even though "caller tune" is the main service keyword.

2. Action-Oriented Training

The chatbot should be trained with more examples containing action pairs such as:

- Activate / Deactivate
- Enable / Disable
- Start / Stop
- Subscribe / Unsubscribe

This helps the system distinguish opposite customer intentions.

3. Semantic Validation

Before executing a telecom operation, the chatbot should validate:

Action + Object + Customer

For example:

DEACTIVATE + CallerTune + Customer

The system should verify that all three components are consistent before executing the request.

4. Negation Detection

The semantic analyzer should specifically detect negative and reversal terms such as:

- Don't

- Disable
- Deactivate
- Cancel
- Stop
- Remove

This reduces errors between opposite intents.

5. Confidence Scoring

The chatbot should calculate a confidence score for each predicted intent. If the confidence is low, it can ask the customer for clarification instead of performing an incorrect operation.

6. Continuous Model Training

Incorrect predictions such as Q2 should be added to the training data. Regular retraining can improve the model's ability to distinguish similar telecom intents.

Final Analysis

The semantic representation system correctly identifies **3 out of 4 intents**, giving an accuracy of **75%**. The major error occurs in Q2, where the system identifies the correct object **CallerTune** but reverses the action from **DEACTIVATE** to **ACTIVATE**. Therefore, improving action recognition, context analysis, negation detection, confidence scoring, and semantic validation can significantly improve the reliability of telecom customer-support chatbots.

CASE STUDY 2: FIRST-ORDER PREDICATE CALCULUS FOR SMART MANUFACTURING

1. Production Data Representation

Given Data

Machine	Status
M1	Active
M2	Active
M3	Maintenance
M4	Active

The production data can be represented using First-Order Predicate Calculus as:

Active(M1)

Active(M2)

Maintenance(M3)

Active(M4)

Assume the production relationships are:

Produces(M1, Gear)

Produces(M2, Wheel)

Produces(M3, Gear)

Produces(M4, Shaft)

Predicate Rules

Rule 1:

$\forall x [\text{Active}(x) \rightarrow \text{Producing}(x)]$

Meaning: If a machine is active, then it is producing.

Rule 2:

$\forall x \forall y [(\text{Produces}(x,y) \wedge \text{Active}(x)) \rightarrow \text{Available}(y)]$

Meaning: If an active machine produces a product, that product is available.

Rule 3:

$\forall x [\text{Maintenance}(x) \rightarrow \neg \text{Producing}(x)]$

Meaning: A machine under maintenance is not producing.

2. Currently Available Products

Step 1: Active Machines

From the production data:

Active(M1)

Active(M2)

Active(M4)

Using:

$\text{Active}(x) \rightarrow \text{Producing}(x)$

We can infer:

Producing(M1)

Producing(M2)

Producing(M4)

Step 2: Product Availability

Given:

Produces(M1, Gear)

Since:

Active(M1)

Therefore:

Available(Gear)

Similarly:

Produces(M2, Wheel)

and:

Active(M2)

Therefore:

Available(Wheel)

For M4:

Produces(M4, Shaft)

and:

Active(M4)

Therefore:

Available(Shaft)

Answer

The currently available products are:

Gear, Wheel, and Shaft

Logical Representation

Available(Gear)

Available(Wheel)

Available(Shaft)

3. Effect of Maintenance on Gear Production

Given

Maintenance(M3)

Production relationship:

Produces(M3, Gear)

According to the rule:

Maintenance(x) $\rightarrow$ $\neg$ Producing(x)

Therefore:

$\neg$ Producing(M3)

Since M3 produces Gear but is under maintenance, M3 cannot currently produce Gear.

However

Another machine is producing Gear:

Produces(M1, Gear)

and:

Active(M1)

Therefore:

Producing(M1)

and:

Available(Gear)

Answer

Maintenance of **M3 affects Gear production capacity**, because M3 cannot produce Gear while under maintenance. However, **Gear is still available** because M1 is active and also produces Gear.

Conclusion

Gear production is partially affected, but not completely stopped.

The system can therefore distinguish between:

Machine-level impact: M3 cannot produce Gear.

Overall product availability: Gear remains available through M1.

4. Effectiveness of Predicate Logic

First-Order Predicate Calculus is useful for industrial decision-support systems because it represents machines, products, statuses, and relationships using formal logical statements.

Advantages

1. Clear Representation

Machine conditions can be represented precisely:

Active(M1)

Maintenance(M3)

This provides a structured representation of plant information.

2. Automated Inference

The system can automatically derive new information from existing facts.

For example:

Active(M1)

and

Active(x) → Producing(x)

allow the system to infer:

Producing(M1)

3. Maintenance Detection

Predicate logic can identify machines that cannot produce:

Maintenance(M3) $\rightarrow$ $\neg$ Producing(M3)

This helps detect production constraints.

4. Product Availability

The system can determine whether a product is available based on the status of the machine producing it.

5. Decision Support

The logical rules can support decisions related to:

- Production planning.
- Machine maintenance.
- Product availability.
- Resource allocation.
- Production disruptions.

Limitations

- Basic predicate logic does not naturally represent uncertainty.
- Real manufacturing environments contain continuously changing data.
- Complex industrial systems may require many rules.
- Sensor errors and incomplete information can affect conclusions.
- Probabilistic or machine-learning techniques may be needed for uncertain situations.

Final Analysis

First-Order Predicate Calculus provides a clear and systematic method for representing manufacturing knowledge and deriving logical conclusions. In this case, it identifies **M1, M2, and M4 as active**, determines that **Gear, Wheel, and Shaft are available**, and shows that **M3's maintenance reduces Gear production capacity without completely stopping Gear availability**. Therefore, predicate logic is effective for rule-based industrial decision support, especially when machine conditions and production relationships can be expressed using clear logical rules.

CASE STUDY 3: WORD SENSE DISAMBIGUATION IN E-COMMERCE SEARCH ENGINES

1. Contextual Sense Analysis

The correct word sense can be identified by examining the **query context** and the **clicked search result**.

User Query	Possible Senses	Clicked Result	Correct Sense
------------	-----------------	----------------	---------------

Apple accessories	Fruit / Technology Brand	iPhone Charger	Technology Brand
Mouse wireless	Animal / Computer Device	Bluetooth Mouse	Computer Device
Java tutorial	Island / Programming Language	Coding Lessons	Programming Language
Python course	Snake / Programming Language	Software Development Training	Programming Language

Analysis

"Apple accessories"

The word "accessories" is associated with electronic products such as chargers, cables, and phone cases. The clicked result is **iPhone Charger**, so "Apple" refers to the **technology brand**.

"Mouse wireless"

The term "wireless" strongly indicates an electronic device. The clicked result is **Bluetooth Mouse**, so "mouse" refers to a **computer device**, not an animal.

"Java tutorial"

The word "tutorial" combined with the clicked result **Coding Lessons** indicates that Java refers to the **programming language**.

"Python course"

The word "course" and the clicked result **Software Development Training** indicate that Python refers to the **programming language**, not the snake.

Final Sense Mapping

Apple → **Technology Brand**

Mouse → **Computer Device**

Java → **Programming Language**

Python → **Programming Language**

2. Semantic Cues for Disambiguation

Semantic cues are words or contextual information that help determine the intended meaning of an ambiguous word.

Apple

Cue: **accessories**

Context: Electronic accessories

Clicked result: **iPhone Charger**

Therefore:

Apple → Technology Brand

Mouse

Cue: **wireless**

Context: Wireless electronic device

Clicked result: **Bluetooth Mouse**

Therefore:

Mouse → Computer Device

Java

Cue: **tutorial**

Context: Programming education

Clicked result: **Coding Lessons**

Therefore:

Java → Programming Language

Python

Cue: **course**

Context: Software development education

Clicked result: **Software Development Training**

Therefore:

Python → Programming Language

Important Semantic Cues

- **Accessories + iPhone Charger** → Technology Brand
- **Wireless + Bluetooth Mouse** → Computer Device
- **Tutorial + Coding Lessons** → Programming Language
- **Course + Software Development** → Programming Language

The system therefore uses **surrounding words, user query context, clicked results, and semantic relationships** to determine the intended sense.

3. Impact of Incorrect Sense Selection

Incorrect Word Sense Disambiguation can significantly reduce the performance of an e-commerce search engine.

Example 1: Apple

If "Apple accessories" is incorrectly interpreted as a fruit-related query, the system may recommend:

- Fruit baskets.
- Apple juice.
- Fruit storage products.

Instead of:

- iPhone chargers.

- AirPods accessories.
- Apple Watch accessories.

This leads to irrelevant search results.

Example 2: Mouse

If "Mouse wireless" is interpreted as an animal, the system may return:

- Animal information.
- Mouse traps.
- Pet products.

Instead of wireless computer mice.

Example 3: Java

If "Java tutorial" is interpreted as an island, the search engine may return travel information instead of programming courses.

Example 4: Python

If "Python course" is interpreted as a snake, the system may recommend wildlife-related content instead of programming courses.

Overall Effects

Incorrect sense selection can cause:

- Irrelevant search results.
- Lower click-through rate.
- Incorrect product recommendations.
- Poor user experience.
- Increased search abandonment.
- Lower customer satisfaction.
- Reduced conversion rate.
- Incorrect personalization.

Therefore, accurate WSD is important for both **search quality and business performance**.

4. Industrial-Scale WSD Strategy

An industrial e-commerce system should use a combination of **NLP, machine learning, user behavior, and contextual information** rather than relying only on fixed dictionary rules.

1. Contextual Embeddings

Use modern language models to represent words based on their surrounding context.

For example:

"Java tutorial"

and

"Java island travel"

would generate different contextual representations for "Java".

2. Query Context Analysis

Analyze nearby words to identify the intended sense.

Examples:

Apple + charger → Technology

Apple + fruit juice → Fruit

Python + programming → Programming Language

Python + snake → Animal

3. Clickstream Analysis

Use historical user behavior such as:

- Clicked products.
- Search results.
- Purchases.
- Search refinements.
- Browsing history.

For example, repeated clicks on **iPhone Chargers** after "Apple accessories" provide strong evidence that Apple refers to the technology brand.

4. Knowledge Graph

Build relationships between entities and concepts.

For example:

Apple → iPhone → Charger → Electronics

Java → Programming → Tutorial → Coding

This helps the search engine identify semantic relationships between query terms and products.

5. Personalized Disambiguation

User history can improve sense selection.

For example, a user frequently searching for programming courses is more likely to mean **Python programming language** when searching for "Python course".

6. Feedback-Based Learning

Search behavior can continuously improve the WSD model.

High clicks and purchases indicate that the selected sense was likely correct, while repeated query reformulation can indicate incorrect interpretation.

7. Confidence-Based Ranking

The system can assign confidence scores to different senses.

For example:

Apple → Technology Brand = 0.95

Apple → Fruit = 0.05

The system can then prioritize results related to the technology brand.

Final Analysis

The correct senses are **Apple → Technology Brand, Mouse → Computer Device, Java → Programming Language, and Python → Programming Language**. Contextual terms such as **"accessories," "wireless," "tutorial," and "course"**, together with clicked results, provide strong semantic evidence for disambiguation. Incorrect sense selection can produce irrelevant results, reduce user satisfaction, and negatively affect conversion rates. For industrial-scale e-commerce systems, combining **contextual NLP models, clickstream behavior, knowledge graphs, personalization, and feedback-based learning** provides a robust strategy for improving WSD and recommendation accuracy.

CASE STUDY 4: SYNTAX-DRIVEN SEMANTIC ANALYSIS IN HEALTHCARE INFORMATION SYSTEMS

1. Syntactic Structure and Semantic Roles

Syntactic parsing identifies the grammatical relationship between words in a sentence. These relationships help the healthcare NLP system determine **who performs an action, who receives it, and what is affected**.

Sentence 1

"Doctor prescribed medicine to patient."

Structure:

Subject → Verb → Object → Recipient

- Doctor → Subject
- prescribed → Verb
- medicine → Object
- patient → Recipient

Semantic roles:

- **Doctor → Agent**
- **Medicine → Theme/Object**
- **Patient → Recipient**

The Doctor performs the action, the Medicine is what is prescribed, and the Patient receives it.

Sentence 2

"Patient reported severe headache."

Structure:

Subject → Verb → Object

- Patient → Subject
- reported → Verb
- severe headache → Object

Semantic roles:

- **Patient → Experiencer**
- **Headache → Symptom**

The Patient experiences/reports the symptom, while the headache represents the medical symptom.

Sentence 3

"Nurse monitored patient continuously."

Structure:

Subject → Verb → Object → Adverb

- Nurse → Subject
- monitored → Verb
- patient → Object
- continuously → Adverb

Semantic roles:

- **Nurse → Agent**
- **Patient → Patient/Theme**
- **continuously → Manner/Time-related modifier**

The Nurse performs the monitoring action on the Patient.

Sentence 4

"Medicine reduced blood pressure."

Structure:

Subject → Verb → Object

- Medicine → Subject
- reduced → Verb
- blood pressure → Object

Semantic roles:

- **Medicine** → **Cause/Instrument**
- **Blood pressure** → **Affected Entity**

The medicine causes a reduction in blood pressure.

2. Evaluation of Assigned Semantic Roles

Given Roles

Doctor → Agent

Medicine → Instrument

Patient → Recipient

Headache → Symptom

Analysis

Doctor → Agent: Appropriate

The doctor performs the action of prescribing medicine. Therefore, **Agent** is the correct semantic role.

Medicine → Instrument: Partially Appropriate

In "Doctor prescribed medicine to patient", medicine is more accurately a **Theme/Object** because it is the thing being prescribed. It is not necessarily an instrument used by the doctor to perform the action.

However, in:

"Medicine reduced blood pressure."

Medicine can be interpreted as the **Cause** or **Instrument** responsible for producing the effect.

Patient → Recipient: Appropriate

In "Doctor prescribed medicine to patient", the patient receives the medicine. Therefore, **Recipient** is appropriate.

Headache → Symptom: Appropriate

A severe headache represents a medical symptom reported by the patient. Therefore, **Symptom** is appropriate.

Corrected Representation

Entity	Given Role	Evaluation	More Appropriate Role
Doctor	Agent	Correct	Agent
Medicine	Instrument	Context-dependent	Theme/Object
Patient	Recipient	Correct in prescription sentence	Recipient
Headache	Symptom	Correct	Symptom

The major issue is the assignment of **Medicine as Instrument** in the prescription sentence. **Theme/Object** is more linguistically accurate there.

3. Errors from Incorrect Parsing

Incorrect syntactic parsing can cause serious problems in healthcare NLP because semantic interpretation depends heavily on grammatical relationships.

Example

Consider:

"Doctor prescribed medicine to patient."

If the system incorrectly identifies the patient as the object performing the action, it could produce:

Patient → Agent

instead of:

Doctor → Agent

This would completely change the meaning of the medical event.

Possible Errors

1. Incorrect Agent Identification

The system may identify the wrong person as performing the medical action.

2. Incorrect Patient Identification

A patient may be incorrectly interpreted as an agent instead of a recipient or experiencer.

3. Incorrect Medical Entity Relationships

The system may incorrectly associate a medicine with the wrong patient.

4. Incorrect Symptom Extraction

A symptom such as "severe headache" could be separated incorrectly or assigned to the wrong entity.

5. Incorrect Treatment Interpretation

The system could misunderstand whether a medicine was prescribed, administered, stopped, or caused an improvement.

6. Incorrect Clinical Decision Support

Incorrect semantic relationships can affect medical summaries, patient records, alerts, and downstream decision-support systems.

Example of Serious Ambiguity

"Patient reported headache after taking medicine."

A poor parser might incorrectly associate "medicine" as the cause of the headache without considering

the temporal relationship expressed by "**after taking**".

Therefore, accurate syntactic and semantic analysis is particularly important in healthcare applications.

4. Methods to Improve Syntax-Driven Semantic Analysis

1. Medical NLP Models

Use healthcare-specific NLP models trained on medical documents. These models are better at understanding terminology such as medicines, symptoms, diseases, and treatments.

2. Dependency Parsing

Use dependency parsing instead of relying only on simple Subject-Verb-Object structures.

Dependency relationships can identify:

- Who performed an action.
- What was affected.
- Who received treatment.
- Relationships between symptoms and patients.

3. Named Entity Recognition

Use medical NER to identify:

- Patients.
- Doctors.
- Medicines.
- Diseases.
- Symptoms.
- Procedures.
- Measurements.

4. Semantic Role Labeling

Apply Semantic Role Labeling to identify roles such as:

- Agent.
- Patient.
- Recipient.
- Theme.
- Cause.
- Experiencer.

This provides a deeper understanding than POS tagging alone.

5. Medical Ontologies

Use medical knowledge bases and ontologies to connect entities and concepts.

For example:

Medicine → Treatment → Disease/Symptom

This helps validate whether extracted relationships are medically reasonable.

6. Context and Negation Detection

The system should recognize terms such as:

- No
- Not
- Without
- Denies

For example:

"Patient has no headache."

The system must not interpret this as the patient having a headache.

7. Temporal Analysis

Healthcare reports frequently contain time-related information.

For example:

"Patient developed fever after surgery."

The system should understand the temporal relationship between surgery and fever.

8. Human Validation

For high-risk clinical applications, important extracted information should be validated by healthcare professionals or through appropriate clinical safeguards.

Final Analysis

Syntactic structure provides the foundation for assigning semantic roles in healthcare NLP. In the given sentences, **Doctor** is correctly identified as an Agent, **Patient** as a Recipient, and **Headache** as a Symptom. However, **Medicine** is better interpreted as a **Theme/Object** in the prescription sentence and as a **Cause/Instrument** when describing its effect on blood pressure. Incorrect parsing can lead to wrong patient-treatment relationships and potentially serious clinical interpretation errors. Therefore, healthcare systems should combine **dependency parsing, medical NER, semantic role labeling, medical ontologies, negation detection, and temporal analysis** to achieve reliable syntax-driven semantic understanding.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Q. No. / Criteria Level	Understanding of Data	Analysis	Application of Concepts	Conclusion	Clarity	Sub. Total	Total %
1							
2							
3							
4							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____